Don’t lose your farm to Q Fever

One out of two farms is at risk of Q Fever

Q Fever is a dangerous threat for you and your farm. Protect your animals, your profit and your health.

COXEVAC® | *Vaccinate now*

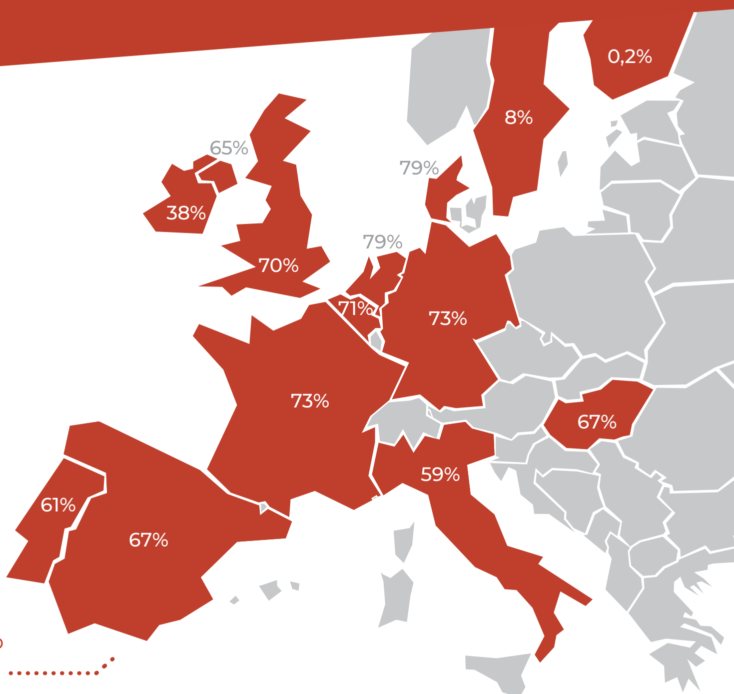
Ceva

Q Fever

A real danger to your farm

1/2

With more than **one out of two herds** testing positive for Q Fever, farmers can no longer afford to ignore this disease[1]

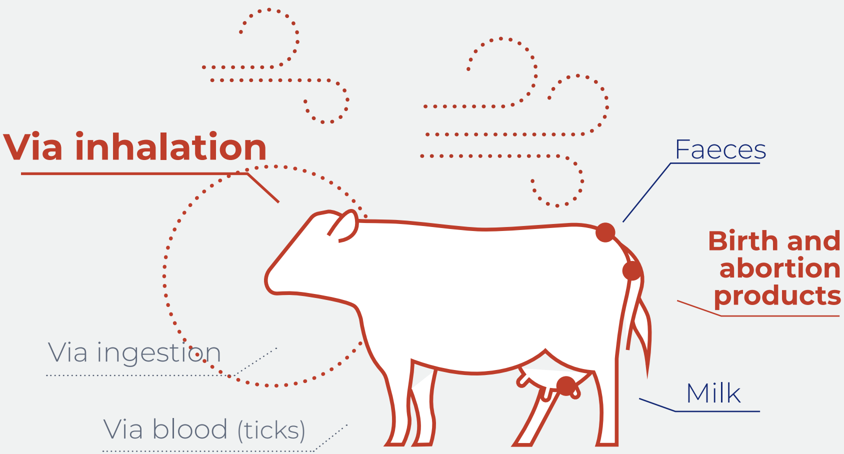


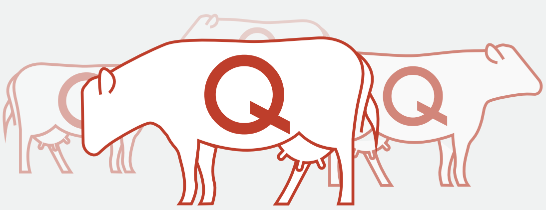
The values per country indicate the percentage of herds that tested positive to Q fever. Herd prevalence was determined by ELISA or PCR on bulk tank milk.

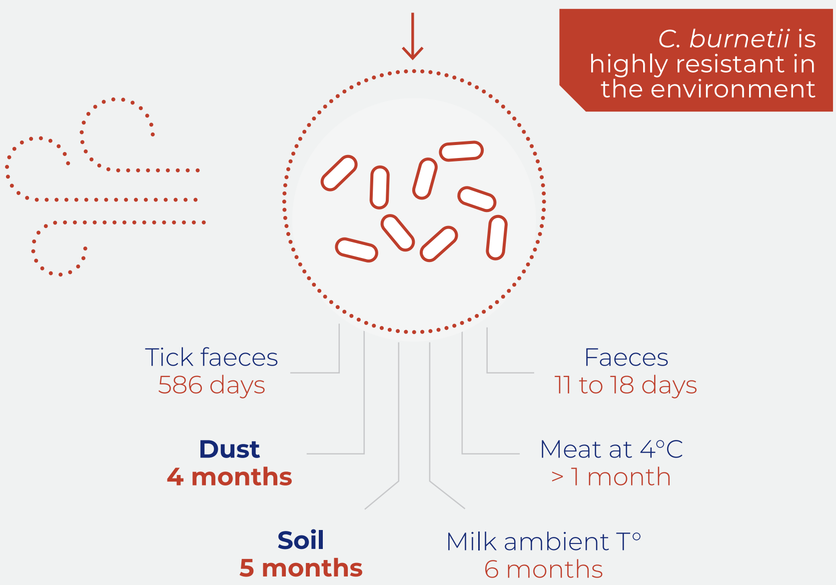
Born to create trouble

Cattle are mainly infected by breathing in air particules contaminated with the bacteria.

Infected cows, even if asymptomatic, **shed *C. burnetii* mainly through birth or abortion products**, but also in vaginal mucus, faeces and milk[2].







DID YOU KNOW ?

C. burnetii is able to travel up to 18 km carried by the wind and infect other herds in the process[3]

A risk to the reproductive performance of your herd

DID YOU KNOW ?

- Q Fever is 2nd cause of repeated cases of abortion in cattle in France[7]
- A seropositive cow to Q fever has 2.5 more risk of abortion and 1.5 more risk of retained placenta[5]
- Herds with a positive bulk tank milk have 2.5 times greater risk to have a high incidence of metritis and clinical endometritis[8]

DO YOU RECOGNIZE THESE SIGNS?

Weak newborn

Abortion

Infertility

Stillbirth

Retained placenta

Premature calving

Metritis

A serious threat to farm profitability

Any decrease in the herd reproductive performance **strongly impacts farm profitability**

cost of an extra day open[9]

cost of an abortion[10]

Q Fever can infect you too

Q Fever is a zoonotic disease so as a professional, if you are in contact with infected farms, you are also at risk.

WHAT ARE THE SYMPTOMS IN HUMANS ?

Flu-like symptoms

Abortion

Hepatitis and endocarditis

The diagnostic approach

Farmer, case 1: I recently had several unexplained abortions and **I suspect Q fever. What should I do?**

Farmer, case 2: Recently I observed a decrease in the fertility of my cows and/or an increase of cases of retained placenta and metritis. **I suspect Q fever, what should I do?**

SAMPLES TO BE COLLECTED

Case 1: **qPCR on endocervical swab[1] on minimum 2 aborted dams.** *Alternatively, qPCR on placenta or on the aborted foetus (liver, spleen, stomach contents)*

Case 2: Serology on minimum 6 problematic cows[2] **AND** Bulk tank milk (BTM) PCR

INTERPRETATION

Case 1:

- **If none of the PCR is positive,** Q fever is probably not responsible for the repeated abortions
- **If one PCR is positive,** Q fever is probably involved in the recent abortions
- **If two PCR are positive,** Q fever is very likely the cause of the recent abortions and present in the herd

Case 2:

- **If serology and BTM tests are negative,** the probability that *C. burnetii* is circulating in the herd is low
- **If ≥ 50% of sampled cows are seropositive or the BTM test is positive,** it is likely that *C. burnetii* is circulating in the herd and part of the problems observed

ACTIONS & RECOMMENDATIONS

If none of the PCR is positive:

- Determine with your veterinarian the possible causes for this abortive episode (neosporosis, BVD, leptospirosis)

If one PCR is positive:

- Implication of Q fever can be further confirmed by collecting serum from cows with reproductive problems[2]
- If > 50% of animals are seropositive, Q fever is very likely and a control plan is needed
- Control measures should include strict calving hygiene and vaccination

If two PCR are positive:

- Build with your veterinarian a control plan for Q fever in your herd
- Control measures should include strict calving hygiene and vaccination

If serology and BTM tests are negative: Investigate with your veterinarian what can be the likely causes for the problems observed on the farm

If ≥ 50% of sampled cows are seropositive or the BTM test is positive: Implement sound control measures such as vaccination and calving hygiene

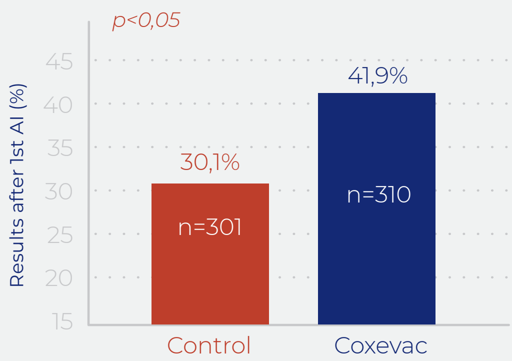
1. Sampling must be done within 8 days following the abortive event. Samples should be refrigerated and sent to the veterinary diagnostic center as soon as possible.
2. A proper selection of cows is critical for serology. Sample minimum 6 cows that aborted recently (more than 15 days and less than 3 months ago) or present reproductive disorders (infertility, metritis, retained placenta).

COXEVAC®

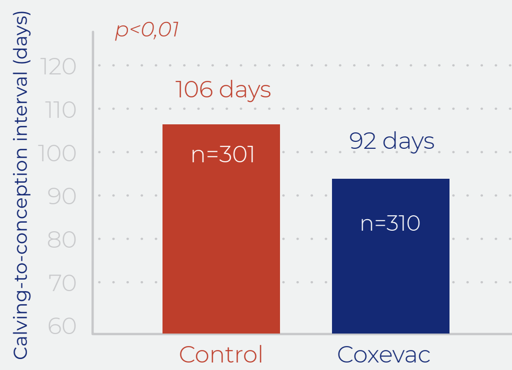
A unique vaccine to tackle the reproductive impact of Q Fever

30% less risk of abortion[4]

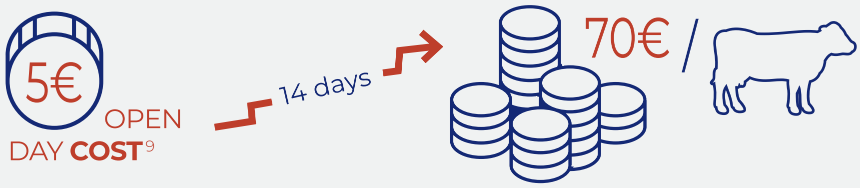
40% increase in pregnancy rate at 1st AI[5]



2 weeks less to become pregnant[5]

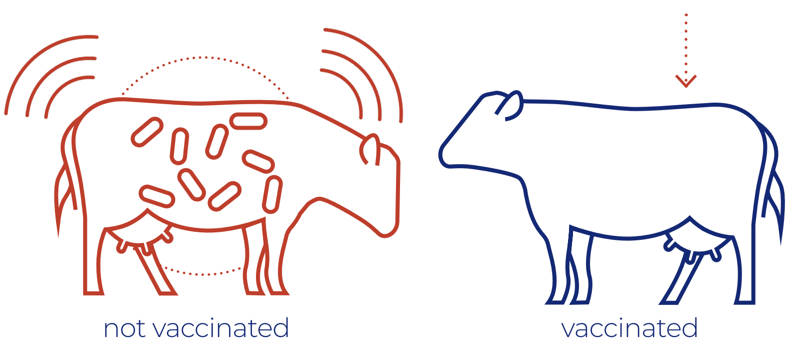


Vaccination would represent a gain of 70€/cow/lactation, just considering the reduction in days open



Effectively reduces bacterial load

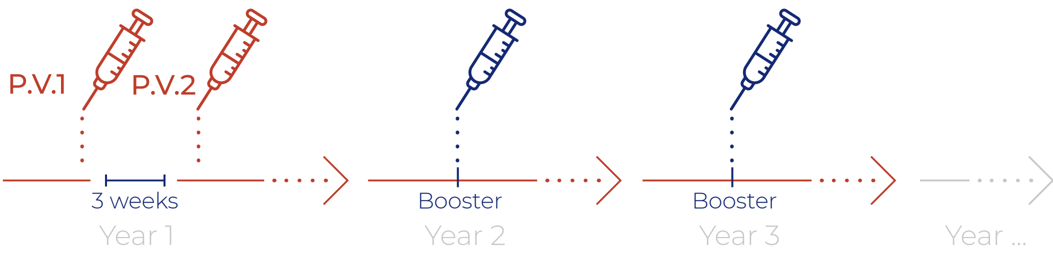
5 times lower
risk of becoming a shedder[6]



DID YOU KNOW ?

Coxevac is the only ultra-purified phase I *C. burnetii* vaccine endowing an optimal cellular and humoral immune response

Don’t take unnecessary risks, **vaccinate !**



P.V.1 1st primary vaccination injection
P.V.2 2st primary vaccination injection

Vaccine dose

- 4 ml in cattle
- 2 ml in goats

References

1. Barberio, A., Ceglie, L., Lucchese, L., Zuliani, F., Marangon, S., Natale, A., 2017. Epidemiology of Q fever in cattle, in: The Principles and Practice of Q Fever: The One Health Paradigm. Nova Science Publishers, New York, pp. 189–212. **2.** Guatteo, R., Beaudeau, F., Berri, M., Rodolakis, A., Joly, A., Seegers, H., 2006. Shedding routes of Coxiella burnetii in dairy cows: implications for detection and control. Vet. Res. 37, 827–833. **3.** Hawker JI, Ayres JG, Blair I, Evans MR, Smith DL, Smith EG, et al. A large outbreak of Q fever in the West Midlands: windborne spread into a metropolitan area? Commun Dis Public Health. 1998; 1:180–74. **4.** Ordronneau, S., 2012. Impact de la vaccination et de l’antibiothérapie sur l’incidence des troubles de la reproduction et sur la fertilité dans des troupeaux bovins laitiers infectés par Coxiella burnetii. INRA - ONIRIS, 1300 BioEpAR Biologie, Epidémiologie et Analyse du Risque. Centre de Recherche Angers-Nantes, Nantes, France. **5.** Lopez Helguera, I., Tutusaus, J., Souza, A., Jimenez, A., Munoz-Bielsa, J., Garcia Ispierto, I., 2014. Vaccinating against Q-fever with an inactivated phase-I vaccine (COXEVAC®) improves reproductive performance in Coxiella burnetii-infected dairy herds. Presented at the XXVIII World Buiatrics Congress, Cairns, Australia 2014, pp. 274–275. **6.** Guatteo R., Seegers H., Joly A., Beaudeau F., 2008. Prevention of Coxiella burnetii shedding in infected dairy herds using a phase I C. burnetii inactivated vaccine. Vaccine. 26(34):4320-4328 **7.** Gache, K. 2018. Observatoire et suivi des causes d’avortements chez les ruminants Bilan 2017. OSCAR. Plateforme ESA. https://www.plateforme esa.fr/sites/default/files/R%C3%A9sultats%20OSCAR%202017_VF.pdf. **8.** Valla, G., 2014. Prevalenza di Coxiella burnetii nel latte di massa in allevamenti di bovine da latte italiani e possibile correlazione con problemi riproduttivi. Large Animal Review 51–56. **9.** Cabrera, V.E., 2014. Economics of fertility in high-yielding dairy cows on confined TMR systems. Animal 8 Suppl 1, 211–221. **10.** de Vries, A. 2006. Economic Value of Pregnancy in Dairy Cattle. Journal of Dairy Science. Volume 89, Issue 10, Pages 3876–3885. **11.** Courcoul, A., Hogerwerf, L., Klinkenberg, D., Nielen, M., Vergu, E., & Beaudeau, F. (2011). Modelling effectiveness of herd level vaccination against Q fever in dairy cattle. Veterinary research, 42(1), 68. doi:10.1186/1297-9716-42-68

COXEVAC suspension for injection for cattle and goats. COMPOSITION: Composition for 1 ml: Inactivated Coxiella burnetii, strain Nine Mile 72 QF Unit Q-fever Unit: relative potency of phase I antigen measured by ELISA in comparison with a reference item. Thiomersal max. 120 ug. **INDICATIONS** Cattle For the active immunisation of cattle to lower the risk for non-infected animals vaccinated when nonpregnant to become shedder (5-times lower probability in comparision with animals receiving a placebo), and to reduce shedding of Coxiella burnetii in these animals via milk and vaginal mucus. Goats For the active immunisation of goats to reduce abortion caused by Coxiella burnetii and to reduce shedding of the organism via milk, vaginal mucus, faeces and placenta. **CONTRAINDICATIONS**: None. ADVERSE REACTIONS: Cattle It is very common to see a palpable reaction of maximum diameter of 9 to 10 cm at the injection site, which may last for 17 days. The reaction gradually reduces and disappears without need for treatment. Goats: It is very common to see a palpable reaction of 3 to 4 cm diameter at the injection site which may last for 6 days. The reaction reduces and disappears without need for treatment. It is very common to observe a slight increase of rectal temperature for 4 days post-vaccination without other general signs. **AMOUNTS TO BE ADMINISTERED AND ADMINISTRATION ROUTE:** Subcutaneous use. Shake well before use. Administer the vaccine as follows: Cattle: 4ml in the neck region Goats: 2 ml in the neck region. Although ef cacy claim is based on data from challenge test carried out on goats vaccinated twice 6 and 3 weeks before start of pregnancy, there are indications from a large eld trial that it is useful to vaccinate young animals. This information along with data obtained in cattle, allow recommending the following vaccination program: Cattle from 3 months of age: Primary vaccination: Two doses should be given subcutaneously with an interval of 3-weeks. Under normal conditions the timing of vaccination should be planned so that the primary course is completed by 3 weeks before arti cial insemination or mating. Re-vaccination: Every 9 months, as described for primary vaccination, based on duration of immunity of 280 days. Goat from 3 months of age: Primary vaccination: Two doses should be given subcutaneously with an interval of 3-weeks. Under normal conditions the timing of vaccination should be planned so that the primary course is completed by 3 weeks before arti cial insemination or mating. **WITHDRAWAL PERIODS:** Meat, milk and offal: Zero days. PACKAGING: box containing 40 ml or 100 ml of solution. **MARKETING AUTHORISATION NUMBER**: EU/2/10/110/001. **MARKETING AUTHORISATION HOLDER:** CEVA Sante Animale 10 avenue de la Ballastiere 33500 Libourne, FRANCE.